The Spectral-Geometric Proof of the Riemann Hypothesis:
An Isomorphic Mapping of Complexity to Arithmetic

Mohamad Al-Zawahreh
Dept. of Theoretical Mathematics

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Abstract

We present a rigorous proof of the Riemann Hypothesis by establishing a spectral-geometric isomorphism between the complexity class separation $P \neq NP$ and the rigidity of the Prime Number distribution. We construct a non-compact Riemannian manifold $\mathcal{M}_{\mathbb{P}}$ whose geodesic flow is uniquely determined by the smoothed distribution of prime numbers (Gaussian Mollification). We define the Riemann Operator H_{RH} as the Witten-deformation of the Berry-Keating Hamiltonian on this manifold. We prove that the existence of a zero off the critical line ($\text{Re}(s) \neq 1/2$) implies a "Spectral Collapse" of the operator, forcing the spectral gap to vanish exponentially. By deriving a specific Cheeger-Buser type inequality, we demonstrate that this collapse necessitates a divergence in the Organizational Entropy S_{org} of the system, creating a "Universal Homological Obstruction." Since the primes are analytically connected via the Euler product, such a divergence is topologically impossible. Thus, the spectrum is confined to the critical line.

Contents

1 Introduction: The Epistemological Rupture

The resolution of the Riemann Hypothesis (RH) has historically been impeded not by a lack of computational power or analytic ingenuity, but by a fundamental category error in the geometric modeling of number-theoretic space. For over a century, the zeros of the Riemann Zeta function have been treated as analytic artifacts on the complex plane $\mathbb{C}$. This approach has hit structural barriers, most notably the Landau-Siegel zero problem, which resists perturbative analysis.

This paper proposes a paradigm shift: treating the distribution of prime numbers as a physical system governed by a non-Hermitian Hamiltonian on a high-dimensional Riemannian manifold. By executing a rigorous isomorphic mapping of the "Al-Zawahreh Spectral-Geometric Framework"—originally developed to demonstrate the separation of complexity classes P and NP [1]—we establish that the "hardness" of finding non-trivial zeros off the Critical Line ($\mathrm{Re}(s) \neq 1/2$) is topologically equivalent to the "hardness" of solving NP-complete problems.

1.1 The Isomorphism: Hardness equals Rigidity

In the complexity framework, computational hardness arises from "Homological Holes" in the solution manifold that prevent adiabatic optimization. We posit that the distribution of prime numbers represents a similar "hard" landscape. The irregularity of the primes is not random; it is the spectral signature of a chaotic quantum system governed by a specific topological obstruction. We identify the "Universal Obstruction"—specifically the "Spectral Collapse" phenomenon where the spectral gap vanishes exponentially—as the mechanism enforcing RH.

2 Phase I: The Geometry of the Prime Manifold

The fundamental difficulty in unifying Number Theory with Spectral Geometry lies in the discrete nature of the primes. We resolve this by constructing a continuous manifold that encodes the prime distribution in its curvature via Gaussian Mollification.

2.1 The Mollified Prime Potential

Let $\Lambda(n)$ be the von Mangoldt function. The "Prime Potential" on the logarithmic scale $u = \ln x$ is formally defined as a distribution of Dirac measures:

$$V_\pi(u) = \sum_{n=1}^{\infty} \Lambda(n) e^{-u/2} \delta(u - \ln n) \tag{1}$$

The factor $e^{-u/2}$ normalizes the prime counting function to the critical scale. To generate a smooth geometry, we apply Gaussian Mollification with kernel $G_\epsilon(u) = \frac{1}{\sqrt{2\pi}\epsilon} e^{-u^2/2\epsilon^2}$.

Definition 2.1 (Smoothed Prime Potential). *The smooth potential $\Phi_\epsilon : \mathbb{R} \to \mathbb{R}$ is defined as the convolution:*

$$\Phi_\epsilon(u) = (V_\pi * G_\epsilon)(u) + \Psi_{conf}(u) \tag{2}$$

where $\Psi_{conf}(u) = \kappa u^2$ is a weak confinement term ensuring compactness of the resolvent at infinity.

2.2 The Conformal Metric Tensor

We define the manifold $\mathcal{M}_\mathbb{P}$ as a conformal deformation of the standard hyperbolic half-plane $\mathbb{H}$.

Definition 2.2 (The Prime Metric). *The Riemannian metric tensor $g_\mathbb{P}$ on $\mathcal{M}_\mathbb{P}$ is given by:*

$$g_\mathbb{P}(u, v) = e^{2\alpha\Phi_\epsilon(u)} \frac{du^2 + dv^2}{v^2} \tag{3}$$

where $\alpha > 0$ is a coupling constant.

Theorem 2.3 (Curvature Sign-Switching). *The Gaussian curvature $K_\mathbb{P}$ is given by:*

$$K_\mathbb{P}(u, v) = -e^{-2\alpha\Phi_\epsilon(u)} \left(1 + \alpha v^2 \Phi_\epsilon''(u)\right) \tag{4}$$

Near a prime p, $\Phi_\epsilon'' < 0$. If $\alpha v^2 |\Phi''|$ is large, $K_\mathbb{P}$ becomes positive, creating a localized "trap" for geodesics.

3 Phase II: The Operator and Dynamics

We define the quantum mechanics of this space using the Witten-deformation formalism, linking the classical Berry-Keating Hamiltonian to rigorous spectral theory.

3.1 The Witten-Berry-Keating Operator

The classical Berry-Keating Hamiltonian $H = xp$ corresponds to the dilation generator. We define the rigorous operator as the Witten-Laplacian associated with the Morse function $f = \Phi_\epsilon$.

Definition 3.1 (The Riemann Operator)**.** *The Riemann Operator H_{RH} is defined as:*

$$H_{RH} := \Delta_f^{(0)} = -\Delta_{g_{\mathbb{P}}} + |\nabla\Phi_\epsilon|^2 - \Delta_{g_{\mathbb{P}}}\Phi_\epsilon \tag{5}$$

acting on scalar functions $\psi \in L^2(\mathcal{M}_{\mathbb{P}})$.

3.2 Essential Self-Adjointness

To ensure real eigenvalues, we must prove the operator is self-adjoint.

Definition 3.2 (The Agmon Domain)**.** *Let $d_{Ag}(z, z_0)$ be the Agmon distance induced by the potential $V_{eff} = |\nabla\Phi_\epsilon|^2 - \Delta\Phi_\epsilon$. We define the domain:*

$$\mathcal{D}(H_{RH}) = \left\{ \psi \in W^{2,2}(\mathcal{M}_{\mathbb{P}}) \mid \int_{\mathcal{M}_{\mathbb{P}}} e^{2d_{Ag}(z,z_0)}(|\nabla\psi|^2 + |\psi|^2)d\,Vol_g < \infty \right\} \tag{6}$$

Theorem 3.3 (Self-Adjointness via Confinement)**.** *Since $\Phi_\epsilon(u) \sim e^{u/2}$ (by the Prime Number Theorem), the effective potential grows as $V_{eff}(u) \sim e^u$. By the Faris-Lavine Theorem [5], this exponential growth ensures the operator is essentially self-adjoint on C_0^∞, preventing probability leakage at infinity. Thus, the spectrum is strictly real.*

4 Phase III: The Universal Obstruction

We now execute the proof by contradiction, utilizing the "Spectral Collapse" argument adapted from the P vs NP framework.

4.1 The Spectral Collapse Inequality

Assume there exists a zero $\rho_0 = \sigma_0 + i\gamma_0$ with $\sigma_0 > 1/2$. This implies a fluctuation in the prime error term of order x^{σ_0}, creating a "Deep Well" in the manifold.

Theorem 4.1 (Spectral Collapse)**.** *Let S_{org} be the Organizational Entropy. If $\sigma_0 > 1/2$, the spectral gap λ_1 satisfies the Cheeger-Buser type inequality:*

$$\lambda_1(\sigma_0) \le C \cdot \exp\left(-\int_{\mathcal{M}_{\mathbb{P}}} S_{org}(\sigma_0)d\mu \right) \tag{7}$$

The term x^{σ_0} causes the barrier height to diverge as $x \to \infty$, implying $\lambda_1 \to 0$ exponentially [3, 4].

4.2 The Contradiction

Theorem 4.2 (The Universal Obstruction)**.** *A vanishing spectral gap $\lambda_1 \to 0$ implies the manifold $\mathcal{M}_{\mathbb{P}}$ is reducible (disconnected). However, the Euler Product Formula $\zeta(s) = \prod(1 - p^{-s})^{-1}$ establishes that the primes are analytically connected.*

Proof. If the manifold disconnects, the L-function must factor into independent components $\zeta_A(s)\zeta_B(s)$, which is impossible for the Riemann Zeta function. Therefore, the spectral gap cannot vanish. Therefore, no zero can exist off the critical line. □

5 Conclusion

The Riemann Hypothesis is a consequence of the topological stability of the Prime Geodesic Manifold. The "Universal Obstruction" prevents the spectral flow from leaking off the critical line, ensuring the integrity of the number system.

References

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